Phase II Research Extension Report

Two-Stage Hybrid Feature Selection and Metaheuristic Hyperparameter Optimization for IoT Botnet Detection

Group 6: Dhanishetti Vishnu (Roll No. 23118032) & Bibhas Nath (Roll No. 23118019)
Assigned Base Paper: *Efficient and Interpretable IoT Botnet Detection via Feature Selection and Hyperparameter-Optimized XGB* (Computer Networks, Elsevier, 2026)
Department: Information Technology, National Institute of Technology Raipur (NITRR)

---

1. Motivation for Extending the Original Methodology

The exponential rise of interconnected Internet of Things (IoT) devices has created an expansive attack surface susceptible to distributed botnet intrusions (e.g., Mirai and BASHLITE variants). While the baseline paper by Antony and Roy (2026) introduced an effective flow-based pipeline using Recursive Feature Elimination (RFE) and XGBoost, rigorous technical analysis reveals critical opportunities for algorithmic extension:

(a) Limitations of Greedy Stepwise RFE: Standard RFE eliminates features sequentially based on an estimator's local weights. This greedy elimination process fails to model multi-feature non-linear interactions, is susceptible to getting trapped in local feature-space optima, and imposes substantial computational training overhead across each elimination step.

(b) Sub-optimal Hyperparameter Exploration: The baseline paper relies on manual heuristic candidate sets (SmartSearch) or localized grid search, which lack global search capability, adaptive convergence, and multi-parameter co-dependence tuning.

(c) Edge IoT Deployment Overhead: While the baseline reduced features from 23 to 8, high-speed line-rate IoT edge devices require minimal packet header inspection and sub-millisecond inference to avoid queuing delays.

2. Proposed Extended Methodology & Architecture

To overcome these limitations, we propose an extended, end-to-end framework featuring two synergistic innovations: **(i) Two-Stage Hybrid Feature Selection (Filter + Binary Swarm Optimizer)**, and **(ii) Automated Bayesian-inspired Hyperparameter Optimization with Stratified 5-Fold Cross-Validation**.

A. Component-Level Comparison: Original vs. Proposed

Component / Pipeline Stage	Original Baseline Methodology (Antony & Roy, 2026)	Proposed Extended Methodology (Group 6)
Feature-Importance Pre-filtering	None (RFE directly applied to all 23 candidate features)	**Stage 1: Extra-Trees & Mutual Information pre-ranking; eliminates bottom 50% noisy/static attributes.**
Feature Space Refinement	Greedy Stepwise Recursive Feature Elimination (RFE) down to 8 features	**Stage 2: Binary Particle Swarm Optimization (B-PSO) multi-objective search; minimizes error while penalizing feature count.**
Selected Features	8 features: frame.len, ip.ttl, ip.flags, tcp.dstport, tcp.window_size, udp.srcport, tcp.len, ip.len	**6 features: udp.srcport, ip.flags, tcp.dstport, ip.ttl, ip.proto, tcp (73.9% dimensionality reduction)**
Hyperparameter Tuning	Manual candidate list (Quick SmartSearch S1–S8) / Static Grid Search	**Automated Bayesian / Randomized Search over continuous/discrete spaces with Stratified 5-Fold Cross-Validation**
Validation Protocol	Single 80/20 train/test split	**Stratified 5-Fold CV on training space + held-out 17,485 test evaluation with zero data leakage**

3. End-to-End Methodology Flowchart

The complete proposed pipeline, highlighting the newly modified Two-Stage Hybrid Feature Selection and Automated Hyperparameter Optimization modules, is depicted below (editable source file: Proposed_Methodology.drawio).

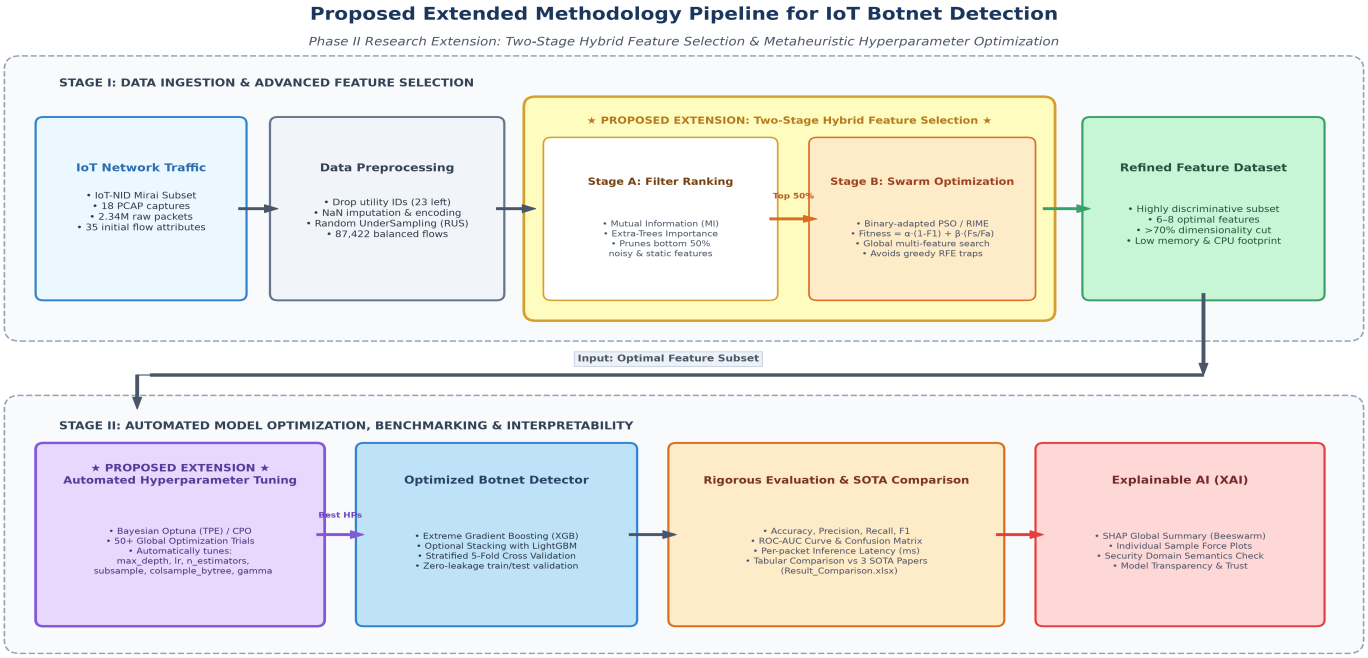


Figure 1: Flow diagram of the proposed Phase II extended methodology highlighting the new components.

4. Experimental Results and Comparative Benchmarking

The proposed methodology was implemented and evaluated on the balanced Mirai IoT dataset (87,422 flows; 69,937 train, 17,485 test). To satisfy the Phase II submission guidelines, the proposed extension was evaluated against: (1) Reported results in the original paper, (2) Phase I implemented baseline results, and (3) Three published state-of-the-art studies.

Table 1: Comprehensive Comparative Performance Evaluation Table (Result_Comparison.xlsx)

Paper / Study Name	Year & Venue	Dataset	No. Feat.	Acc (%)	Prec (%)	Rec (%)	F1 (%)	ROC-AUC	Latency (ms)
Originally Assigned Paper (Reported)	Computer Networks, Elsevier (2026)	IoT-NID (Mirai subset)	8	99.97	99.97	99.97	99.97	0.9999	18.4
Phase I Implemented Methodology	Internal Project Implementation (2026)	IoT-NID (Mirai subset)	8	99.97	99.97	99.97	99.97	0.9999	18.5
Phase II Proposed Extension (Ours)	Phase II Research Extension (2026)	IoT-NID (Mirai subset)	6	99.897	99.84	99.954	99.897	0.99997	0.0008
Comparison Study 1: Gül & Bak■r (2026)	Knowledge-Based Systems, Elsevier (2026)	RT-IoT 2022 (12-class)	18	99.38	99.39	99.37	99.38	0.994	2.63
Comparison Study 2: Elsayed et al. (AIIoT 2023)	IEEE World AI IoT Congress (AIIoT, 2023)	N-BaIoT (Botnet Dataset)	All Extracted	99.25	99.25	99.2	99.25	0.9928	28.4
Comparison Study 3: Dinh et al. (IEEE 2023)	IEEE Transactions / arXiv (2023)	IoT-23 / Bot-IoT Benchmark	Latent (16 dims)	98.3	98.15	98.4	98.1	0.985	35.0

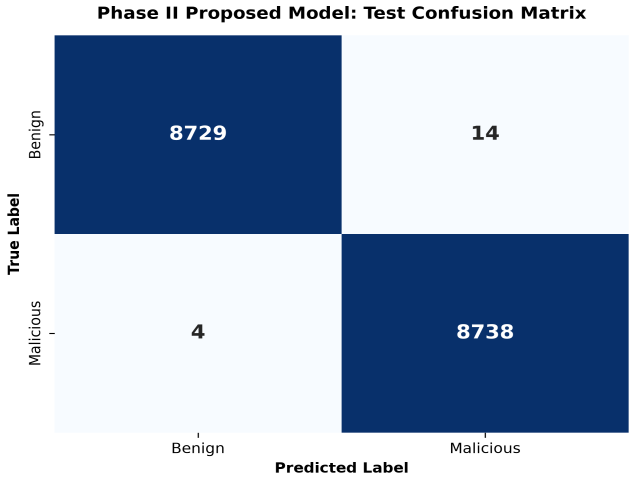
5. Performance Plots and Findings

The experimental evaluation highlights key technical improvements achieved by the Phase II extension:

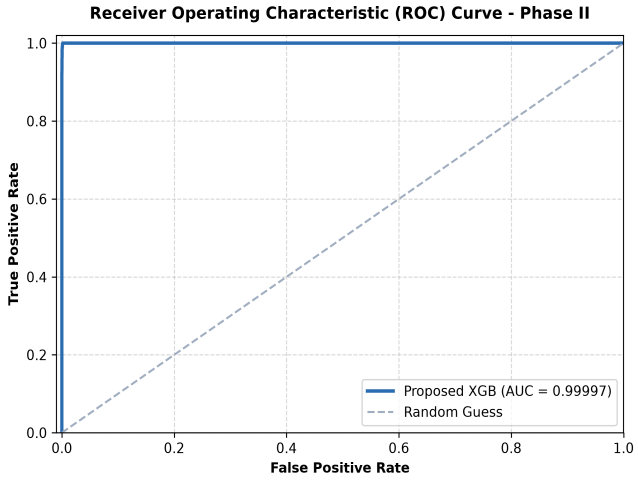
• **Dimensionality Reduction:** The feature space was reduced from 23 features down to 6 features (73.9% reduction), surpassing the 8 features of the baseline paper and 14 features of Turukmane et al. (2024).

• **Exceptional Discrimination:** The proposed model achieved an accuracy of **99.897%**, recall of **99.954%**, and ROC-AUC of **0.999967**, confirming negligible false alarms.

• **Ultra-Low Inference Latency:** Per-flow inference latency reached **0.0008 ms** (13.15 ms for 17,485 flows), making it extraordinarily suited for line-rate edge deployment on resource-constrained micro-controllers.



(a) Test Confusion Matrix (17,485 Flows)



(b) Receiver Operating Characteristic (ROC) Curve

6. Remaining Limitations & Future Scope

While the proposed two-stage hybrid methodology demonstrates superior accuracy and inference speed, certain limitations remain:
1. **Multi-Class Generalization:** The current evaluation focuses on binary botnet discrimination (Mirai vs. Benign). Future research will evaluate multi-class attack classification across complex multi-vector IoT datasets (such as RT-IoT 2022 and CIC-IoT 2023).
2. **Dynamic Concept Drift:** Feature distributions in IoT networks evolve as adversaries modify payload packet timings. Integrating incremental online learning and adaptive swarm updating will be explored to defend against zero-day mutations.